

Lecture 3

Bayes' theorem and Bayesian learning

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(Edited from the lecture notes by Brian J. Reich, NC State)



Bayes' Theorem

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$$\theta = \begin{cases} 0 & \text{you have seasonal flu,} \\ 1 & \text{you have contracted Covid 19.} \end{cases}$$

- ▶ Then, you went to the health center and got tested for Covid 19.
- ▶ The data is Y , defined as

$$Y = \begin{cases} 0 & \text{test is negative,} \\ 1 & \text{test is positive.} \end{cases}$$

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- ▶ Since Y is binary, we use a Bernoulli PMF for the likelihood.
- ▶ We must specify the likelihood for both $\theta = 0$ and $\theta = 1$.
- ▶ $\text{Prob}(Y = 1|\theta = 0) = q_0$ is the false positive rate.
- ▶ $\text{Prob}(Y = 1|\theta = 1) = q_1$ is the true positive rate.
- ▶ How might we select q_0 and q_1 ?

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- ▶ $\text{Prob}(\theta = 1) = p$ is the population prevalence of Covid 19.
- ▶ How might we select p ?
- ▶ Several factors involved: current positivity rate at IIT-K, if anyone tested positive with whom you came in close contact, etc.

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- ▶ This is Bayes' theorem: obtaining $p(\theta|y)$ from $f(y|\theta)$ and $\pi(\theta)$.

Bayes' Theorem

- ▶ In Bayesian statistics, we select the prior, $\pi(\theta)$, and the likelihood, $f(y|\theta)$.
- ▶ Based on these two pieces of information, our main goal is to compute the posterior $p(\theta|y)$.
- ▶ Bayes' theorem is the mathematical formula to convert the likelihood and prior to the posterior.

- ▶ Bayes theorem:

$$p(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{m(y)}.$$

- ▶ This holds for discrete (PMF) and continuous (PDF) cases.

Bayes' theorem

- ▶ Bayes theorem in math:

$$p(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{m(y)}.$$

- ▶ Bayes theorem in words:

$$p(\theta|y) = \frac{\text{Likelihood} * \text{Prior}}{\text{marginal distribution of } Y}.$$

- ▶ As in the formula for a conditional distribution, $m(y)$ is just the normalizing constant required so that $\int p(\theta|y)d\theta = 1$.
- ▶ Most of the time $m(y)$ can be ignored because it doesn't depend on θ and the objective is to study the posterior of θ .

Cricket example

- ▶ Team India plays half its games at home, wins 70% of its home games, and 40% of its away games. Given that the team wins a game, what's the probability it was a home game?

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- ▶ Team India plays half its games at home, wins 70% of its home games, and 40% of its away games. Given that the team wins a game, what's the probability it was a home game?
- ▶ Let θ be the parameter of interest with

$$\theta = \begin{cases} 1 & \text{if it was a home match} \\ 0 & \text{if it was an away match} \end{cases}$$

- ▶ The data is Y , defined as

$$Y = \begin{cases} 1 & \text{if the team wins} \\ 0 & \text{if the team loses} \end{cases}$$

Cricket example

- ▶ Here, $\text{Prob}(\theta = 0) = \text{Prob}(\theta = 1) = 0.5$.
- ▶ Also, $\text{Prob}(Y = 1|\theta = 1) = 0.7$ and $\text{Prob}(Y = 1|\theta = 0) = 0.4$.



$$\begin{aligned} & \text{Prob}(\theta = 1|Y = 1) \\ = & \frac{\text{Prob}(\theta = 1, Y = 1)}{\text{Prob}(Y = 1)} \\ = & \frac{\text{Prob}(Y = 1|\theta = 1)\text{Prob}(\theta = 1)}{\text{Prob}(Y = 1|\theta = 1)\text{Prob}(\theta = 1) + \text{Prob}(Y = 1|\theta = 0)\text{Prob}(\theta = 0)} \\ = & \frac{0.7 \times 0.5}{0.7 \times 0.5 + 0.4 \times 0.5} \\ = & \frac{7}{11}. \end{aligned}$$

Robins example

- ▶ Let X be the number of robins in the forest
- ▶ Let Y be the number of robins we observe
- ▶ Uniform prior: $\text{Prob}(X = x) = 1/20$ for $x \in \{0, \dots, 19\}$
- ▶ Likelihood: $Y|\theta \sim \text{Binomial}(X, 0.2)$ (0.2 is the detection probability)
- ▶ Given that we do not observe any birds, what is the probability that no birds are in the forest?
- ▶ Intuitively, how would this change if the prior was $\text{Prob}(X = x) = 1/100$ for $x \in \{0, \dots, 99\}$?
- ▶ Intuitively, how would this change if the detection probability increased from 0.2 to 0.9?

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$$\begin{aligned} & \text{Prob}(X = 0 | Y = 0) \\ = & \frac{\text{Prob}(Y = 0 | X = 0) \text{Prob}(X = 0)}{\sum_{x=0}^{19} \text{Prob}(Y = 0 | X = x) \text{Prob}(X = x)} \\ = & \frac{1 \times 1/20}{\sum_{x=0}^{19} \binom{x}{0} 0.2^0 0.8^x \times 1/20} \\ = & \frac{1}{\sum_{x=0}^{19} 0.8^x} = 0.2023. \end{aligned}$$

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- ▶ So, the probability decreases. Why this is so?

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- ▶ So, the probability decreases. Why this is so?
- ▶ The prior probability of $\text{Prob}(X = 0)$ is smaller here.

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- ▶ So, the probability increases. Why this is so?
- ▶ With better detection probability, we are more confident in our data.

Bayesian learning

- ▶ Bayesians quantify uncertainty about fixed but unknown parameters by treating them as random variables.
- ▶ This requires that we set a **prior distribution** $\pi(\theta)$ to summarize uncertainty before observing the data.
- ▶ The distribution of the observed data given the model parameters is the **likelihood function**, $f(Y|\theta)$.
- ▶ The likelihood function is the most important piece of a Bayesian analysis because it links the data and parameters.

Bayesian learning

- ▶ The **posterior distribution** $p(\theta|Y)$ summarizes uncertainty about the parameters given the prior and data.
- ▶ The reduction in uncertainty from prior to posterior represents **Bayesian learning**.
- ▶ **Bayes' Theorem** (Bayes' Rule) converts the likelihood and prior to the posterior.
- ▶ Bayes' Theorem:

$$p(\theta|Y) = \frac{f(Y|\theta)\pi(\theta)}{m(Y)}$$

where $m(Y) = \int f(Y|\theta)\pi(\theta)d\theta$ is the marginal distribution of the data and can usually be ignored.

How to select the prior?

- ▶ There is no “true” or “correct” prior.
- ▶ In some cases expert opinion or similar studies can be used to specify an informative prior.
- ▶ It would be a waste to discard this information.
- ▶ If prior information is unavailable, then the prior should be non-informative or weakly informative.
- ▶ The prior is best viewed as an initial value to a statistical procedure.

How to select the prior?

- ▶ As we will see later, as Bayesian learning continues, and further data are collected, the posterior concentrates around the true value for any reasonable prior.
- ▶ However, in a finite sample the prior can have some effect.
- ▶ We will study several systematic ways to select priors.
- ▶ However, there is inherent **subjectivity** to selecting the prior.
- ▶ That is, different analysts may pick different priors and thus have different results.

How to select the likelihood?

- ▶ The likelihood is the same as in a Frequentist analysis.
- ▶ For example, in a linear regression analysis, we might say

$$Y_i | \beta, \sigma^2. \sim \text{Normal} \left(\sum_{j=1}^p X_{ij} \beta_j, \sigma^2 \right).$$

- ▶ Is there “true” or “correct” likelihood?
- ▶ Is specification of the likelihood subjective like the prior?

How to select the likelihood?

Subjective decisions required to specify the likelihood:

- ▶ The errors are independent.
- ▶ $\log(Y_i)$ follows a normal distribution.
- ▶ The mean is a function of X_{i1} , X_{i3} , and X_{i3}^2 .
- ▶ Y_{43} is an outlier and removed.
- ▶ Many, many more.

Opinions about subjectivity

- ▶ Perhaps we should aspire to objectivity, but in most real-life analyses we are forced to accept some subjectivity.
- ▶ A notable exception is a tightly controlled experiment, although even this is debatable.
- ▶ However, not all subjective decisions (assumptions) are equal.
- ▶ If readers disagree with your assumptions they will reject your findings.
- ▶ It is your job to justify your assumptions theoretically and empirically.
- ▶ Determining the sensitivity to key assumptions is an important step.

Thank you!